Math 110 — Week 7

Diagonalizability and the minimal polynomial (Notes Ch. 12–13, LADR 5C, 5D, 5B, HW7)

§1 Facts used from earlier weeks

V is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Shorthand: "S kills v" means $Sv = 0$, and "q kills T" means $q(T) = 0$. From Week 6: eigenvalues, eigenvectors and eigenspaces $E(\lambda, T) = \operatorname{null}(T - \lambda I)$, and invariant subspaces.

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

Why.

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of V. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: T of any vector is a combination of the Tu_i (which are 0) and the Tv_j.
3. Independent: if $\sum c_j Tv_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the u_i, and independence of the basis of V gives all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**

Why. $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n$ (T1) $\iff \operatorname{range} T = V$ (T3).

(T3) **Dimension shortcuts.**

(a) An independent list of n vectors is a basis.

(b) A subspace of dimension n is all of V.

(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Why.

1. (a), (b): an independent list extends to a basis, and every basis has length n.
2. (c): a basis of U followed by a basis of W is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length n, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis x_i, u_j of U and, separately, to a basis x_i, w_l of W. Together the x_i, u_j, w_l span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ (x, u, w combinations of the x_i, u_j, w_l), then $w = -x - u \in U \cap W$, so w is a combination of the x_i. The basis x_i, w_l of W is independent, so $w = 0$. Then $x + u = 0$, and independence of x_i, u_j gives $x = u = 0$.

(T4) **Products of operators.** If S, T are injective, so is ST. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

Why. $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in T commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are T-invariant.

Why. Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda) = 0 \iff (z-\lambda) \mid p$.

Why. Divide p by $z-\lambda$: the remainder is a constant, and evaluating at λ shows it equals $p(\lambda)$.

(T7) **Eigenvalues.**

(a) λ is an eigenvalue $\iff$ $T - \lambda I$ is not injective.

(b) If $Tv = \lambda v$, then $p(T)v = p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

Why.

1. (a) is the definition: $Tv = \lambda v$ with $v \neq 0$ means $(T-\lambda I)v = 0$ with $v \neq 0$.
2. (b): $T^k v = \lambda^k v$ by induction; add up the terms of p.
3. (d): take $v \neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v = 0$ for some $p \neq 0$; p is not constant since $v \neq 0$.
4. Factor $p = c(z-\lambda_1)\cdots(z-\lambda_m)$ (T6). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v \neq 0$ to 0, so some $T - \lambda_j I$ is not injective (T4).

(T8) **Block diagonal matrices.** If $V = U \oplus W$ with both T-invariant, then in a basis of U followed by one of W, $\mathcal{M}(T) = \left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T = \operatorname{null} T|_U \oplus \operatorname{null} T|_W$.

Why. $Tu \in U$ is a combination of the U-basis only, so its column is 0 in the W-rows (same for W). If $T(u+w) = 0$, then $Tu = -Tw \in U \cap W = \{0\}$.

(T9) **Change of basis.** If column j of S holds the B-coordinates of the j-th vector of B', then $\mathcal{M}_{B'}(T) = S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.

Why. S converts B'-coordinates to B-coordinates: apply S, then $\mathcal{M}_B(T)$, then convert back with S^{-1}. Every invertible S arises this way, since its columns form a basis.

(T10) **Quotient operator.** If U is T-invariant, then $(T/U)(v+U) = Tv + U$ is a well-defined operator on V/U, $\dim V/U = \dim V - \dim U$, and $p(T/U)(v+U) = p(T)v + U$.

Why.

1. Well defined: if $v + U = v' + U$, then $v - v' \in U$, so $Tv - Tv' \in U$ (U is invariant).
2. Dimension: $v \mapsto v + U$ maps V onto V/U with null space U; apply (T1).
3. $(T/U)^k(v+U) = T^k v + U$ by induction; add up the terms of p.

§2 This week

§2.1 Upper-triangular matrices (Ch. 12.1)

The question. For $A = \left[\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right]$ over $\mathbb{C}$, every eigenvector is a multiple of $(1,1)$, so there is no eigenbasis. Is there a basis in which the matrix of A is upper-triangular, and can the eigenvalue then be read off without solving an equation?

Definition (Def. 33)**.** A square matrix $A = (a_{ij})$ is **upper-triangular** if $a_{ij} = 0$ whenever $i > j$. Its **diagonal** is $a_{11}, \ldots, a_{nn}$. It is **diagonal** if every entry off the diagonal is 0.

Example 53. The matrix

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

is upper-triangular with diagonal $1, 5, 8, 10$.

Proposition (Prop. 59 / LADR 5.39)

Let $v_1, \dots, v_n$ be a basis of V. The following are equivalent:

(i) the matrix of T in $v_1, \dots, v_n$ is upper-triangular;

(ii) $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each j;

(iii) $\operatorname{span}(v_1, \dots, v_j)$ is invariant under T for each j.

The idea

Column j of the matrix lists the coordinates of Tv_j: $Tv_j = a_{1j}v_1 + \cdots + a_{nj}v_n$. "Upper-triangular" says the entries of column j below row j are 0, which is the statement that Tv_j uses only $v_1, \dots, v_j$. So the proposition only translates a statement about zeros in a matrix into a statement about where basis vectors are sent.

For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $Ne_1 = 0$ and $Ne_2 = e_1$. Each basis vector is sent to a combination of itself and earlier ones, never to a later one.

Proof. 1. (i)⇒(ii): $Tv_j = a_{1j}v_1 + \cdots + a_{nj}v_n$, and $a_{ij} = 0$ for $i > j$, so $Tv_j = a_{1j}v_1 + \cdots + a_{jj}v_j$.

2. (ii)⇒(iii): for $i \le j$, $Tv_i \in \operatorname{span}(v_1, \dots, v_i) \subseteq \operatorname{span}(v_1, \dots, v_j)$. So T sends each of $v_1, \dots, v_j$, and hence their span, into $\operatorname{span}(v_1, \dots, v_j)$.

3. (iii)⇒(i): $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$, so in $Tv_j = a_{1j}v_1 + \cdots + a_{nj}v_n$ the coefficients $a_{j+1,j}, \dots, a_{nj}$ are 0 (coordinates in a basis are unique).

□

Proposition (Prop. 60 / LADR 5.41)

If T has an upper-triangular matrix in some basis, the eigenvalues of T are exactly the diagonal entries of that matrix.

The idea

λ is an eigenvalue exactly when $T - \lambda I$ is not injective (T7). The matrix of $T - \lambda I$ is again upper-triangular, with diagonal entries $\lambda_k - \lambda$. So the question is: when is an upper-triangular matrix injective?

Try $\left(\begin{smallmatrix} 2 & 5 \\ 0 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = 0$. The bottom row says $3y = 0$, so $y = 0$; then the top row says $2x = 0$, so $x = 0$. Each row, read from the bottom up, forces one more coordinate to be 0, as long as the diagonal entry in that row is not 0. So a nonzero diagonal gives an injective matrix.

If instead some diagonal entry is 0, say in position j, then v_j is sent into $\operatorname{span}(v_1, \dots, v_{j-1})$. So j independent vectors $v_1, \dots, v_j$ are all sent into a space of dimension $j - 1$, and that cannot be injective.

Proof. Let $\lambda_1, \dots, \lambda_n$ be the diagonal entries, in the basis $v_1, \dots, v_n$. Then $\mathcal{M}(T - \lambda I)$ is upper-triangular with diagonal entries $\lambda_k - \lambda$.

1. *If $\lambda \neq \lambda_k$ for all k, then $T - \lambda I$ is injective.* Suppose $(T - \lambda I)v = 0$ with $v = c_1v_1 + \cdots + c_nv_n$. Only $(T - \lambda I)v_n$ has a v_n-component, and it is $(\lambda_n - \lambda)v_n$. So the v_n-coefficient of $(T - \lambda I)v$ is $(\lambda_n - \lambda)c_n$, which must be 0; hence $c_n = 0$.

2. Downward induction: suppose $c_n = \cdots = c_{k+1} = 0$, so $v = c_1v_1 + \cdots + c_kv_k$. For $i < k$, $(T - \lambda I)v_i \in \operatorname{span}(v_1, \dots, v_i)$ has no v_k-component, and $(T - \lambda I)v_k$ has the v_k-component $(\lambda_k - \lambda)v_k$. So the v_k-coefficient of $(T - \lambda I)v = 0$ is $(\lambda_k - \lambda)c_k$, and $c_k = 0$. Hence every $c_k = 0$, and λ is not an eigenvalue.

3. *If $\lambda = \lambda_j$, then $T - \lambda I$ is not injective.* For $i < j$, $(T - \lambda I)v_i \in \operatorname{span}(v_1, \ldots, v_i)$ (Prop. 59). And $(T - \lambda I)v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$, because its v_j-coefficient is $\lambda_j - \lambda = 0$.
4. So $T - \lambda I$ maps the j-dimensional space $\operatorname{span}(v_1, \ldots, v_j)$ into the $(j-1)$-dimensional space $\operatorname{span}(v_1, \ldots, v_{j-1})$. By (T1) this restriction has a nonzero null space. So λ is an eigenvalue.

□

Example 54. Differentiation D on $\mathcal{P}_3(\mathbb{R})$ in the basis $1, x, x^2, x^3$ has matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

since $Dx^k = kx^{k-1}$. It is upper-triangular with diagonal $0, 0, 0, 0$, so the only eigenvalue is 0. Each basis vector is sent to a multiple of the previous one: $x^3 \mapsto 3x^2 \mapsto 6x \mapsto 6 \mapsto 0$.

$$\underset{\text{degree 3}}{x^3} \xrightarrow{D} \underset{\text{degree 2}}{3x^2} \xrightarrow{D} \underset{\text{degree 1}}{6x} \xrightarrow{D} \underset{\text{degree 0}}{6} \xrightarrow{D} 0$$

Each application of D lowers the degree by one, so $D^4 = 0$ on $\mathcal{P}_3(\mathbb{R})$ and 0 is the only eigenvalue. Read backward, the chain is the basis $1, x, x^2, x^3$ up to scaling, which is why the matrix has its nonzero entries just above the diagonal.

Theorem (Prop. 61 / LADR 5.47)

If V is a finite-dimensional complex vector space, every $T \in \mathcal{L}(V)$ has an upper-triangular matrix in some basis.

The idea

By Prop. 59 we need a basis in which each Tv_k lies in the span of v_k and the earlier vectors. Induction suggests: find a smaller invariant subspace U, take such a basis of U (induction), and then add more vectors. The added vectors v must satisfy $Tv \in U + \operatorname{span}(v)$.

Which U makes that automatic? Write $Tv = (T - \lambda I)v + \lambda v$. The second term is a multiple of v. The first term lies in $\operatorname{range}(T - \lambda I)$, whatever v is. So take $U = \operatorname{range}(T - \lambda I)$: then *every* vector v satisfies $Tv \in U + \operatorname{span}(v)$.

For induction U has to be smaller than V, i.e. $T - \lambda I$ must not be surjective, which for operators is the same as not injective (T2). That is exactly the condition that λ is an eigenvalue. Over $\mathbb{C}$ an eigenvalue exists (T7); this is the only place $\mathbb{C}$ is used.

Proof. Induction on $n = \dim V$; for $n = 1$ every matrix is upper-triangular.

1. Take an eigenvalue λ of T (T7) and let $U = \operatorname{range}(T - \lambda I)$. U is invariant under T (T5). $T - \lambda I$ is not injective, so it is not surjective (T2), so $\dim U < n$.
2. By induction there is a basis $u_1, \ldots, u_m$ of U in which $T|_U$ is upper-triangular: $Tu_i \in \operatorname{span}(u_1, \ldots, u_i)$.
3. Extend to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of V. Then $Tv_j = (T - \lambda I)v_j + \lambda v_j$, with $(T - \lambda I)v_j \in U = \operatorname{span}(u_1, \ldots, u_m)$. So $Tv_j \in \operatorname{span}(u_1, \ldots, u_m, v_1, \ldots, v_j)$.
4. By Prop. 59 (ii)⇒(i), the matrix of T in this basis is upper-triangular.

□

Over $\mathbb{R}$ this can fail. In an upper-triangular basis, $Tv_1 \in \operatorname{span}(v_1)$, so v_1 is an eigenvector. The rotation $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ of $\mathbb{R}^2$ has no eigenvector, so it has no upper-triangular matrix over $\mathbb{R}$. Which real operators do have one is answered by LADR 5.44 in Section 2.4.

The question answered. Yes. $A(1, 1) = (2, 2)$ and $Ae_2 = (1, 3) = (1, 1) + 2e_2$, so in the basis $(1, 1), e_2$ the matrix is $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$, and the only eigenvalue 2 is on the diagonal (Prop. 60).

Shortcuts

- **Triangular matrix: the eigenvalues are the diagonal entries.** Reason: Prop. 60. Condition: the matrix is triangular (upper or lower; a lower-triangular matrix is upper-triangular in the reversed basis). Instance:
$$\begin{pmatrix} 2 & 5 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 2 \end{pmatrix} \text{ has the eigenvalues } 2, 3.$$
- **An upper-triangular basis starts with an eigenvector.** Reason: $Tv_1 \in \operatorname{span}(v_1)$ (Prop. 59). Instance: in the question, $v_1 = (1, 1)$ is the eigenvector.

Traps

- The eigenvalues are the diagonal entries only for a *triangular* matrix: $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has diagonal $0, 0$ but eigenvalues ± 1.
- Over $\mathbb{R}$ an operator may have no upper-triangular matrix at all: the rotation by $90°$.

§2.2 Diagonalizable operators (Ch. 12.2)

The question. When is there a basis in which T only rescales each basis vector?

- $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$: $A(1,1) = 3(1,1)$ and $A(1,-1) = (1,-1)$. In the basis $(1,1), (1,-1)$ the matrix is $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.
- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: every eigenvector is a multiple of e_1, so there are not enough eigenvectors for a basis.

Definition (Def. 34 / LADR 5.50)**.** T is **diagonalizable** if its matrix in some basis of V is diagonal. A basis of V consisting of eigenvectors of T is an **eigenbasis**. The **geometric multiplicity** of λ is $\dim E(\lambda, T) = \dim \operatorname{null}(T - \lambda I)$; it is 0 when λ is not an eigenvalue.

Theorem (Prop. 62 / LADR 5.55)

Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of T. The following are equivalent:

(i) T is diagonalizable;

(ii) V has a basis of eigenvectors of T;

(iii) $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$;

(iv) $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

The idea

(i) and (ii) say the same thing: column j of the matrix is the coordinates of Tv_j, and that column is 0 except for the entry λ in row j exactly when $Tv_j = \lambda v_j$.

The content is in (iii) and (iv). The eigenspaces for different eigenvalues never overlap: their sum is always direct, because eigenvectors for distinct eigenvalues are independent (T7). So eigenspaces never give *too many* vectors. The only way to fail is to give *too few*, and that is a question of counting dimensions. That is why the single number in (iv) decides diagonalizability.

For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: the only eigenvalue is 0 and $\dim E(0, N) = 1 < 2$.

Proof. 1. (i) $\Longleftrightarrow$ (ii): the matrix in $v_1, \ldots, v_n$ is diagonal with entries $\mu_1, \ldots, \mu_n$ exactly when $Tv_j = \mu_j v_j$ for every j, i.e. when every v_j is an eigenvector.

2. *The sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is always direct.* Suppose $u_1 + \cdots + u_m = 0$ with $u_i \in E(\lambda_i, T)$. The nonzero u_i are eigenvectors for distinct eigenvalues, so they are independent (T7); a sum of independent vectors is not 0. So there are no nonzero u_i.

3. (ii)⇒(iii): each vector of an eigenbasis lies in some $E(\lambda_i, T)$, so the sum of the eigenspaces contains a basis of V and equals V. It is direct by step 2.
4. (iii)⇒(iv): the dimension of a direct sum is the sum of the dimensions.
5. (iv)⇒(iii): by step 2 the sum of the eigenspaces is direct, so its dimension is $\sum \dim E(\lambda_i, T) = \dim V$. A subspace of full dimension is V (T3).
6. (iii)⇒(ii): choose a basis of each $E(\lambda_i, T)$. Because the sum is direct, together they form a basis of V, and every vector in it is an eigenvector.

□

Example 55: an operator that is not diagonalizable. For every $n \geq 2$ there is an operator on V that is not diagonalizable (and for $n \leq 1$ every operator is diagonal in every basis). Example: let $Tv_1 = 0$ and $Tv_k = v_{k-1}$ for $k \geq 2$, with $n \geq 2$. The matrix is upper-triangular with zero diagonal, so the only eigenvalue is 0 (Prop. 60). $\operatorname{null} T = \operatorname{span}(v_1)$: if $T(\sum c_k v_k) = \sum_{k \geq 2} c_k v_{k-1} = 0$, then $c_2 = \cdots = c_n = 0$. So $\sum_\lambda \dim E(\lambda, T) = 1 \neq n$, and T is not diagonalizable by (iv).

Proposition (Prop. 63 / LADR 5.58)

If T has $n = \dim V$ distinct eigenvalues, then T is diagonalizable.

Proof. 1. Choose an eigenvector v_i for each of the n eigenvalues λ_i.

2. They are independent (T7), and there are n of them, so they form a basis (T3).
3. Every vector of this basis is an eigenvector, so it is an eigenbasis, and T is diagonalizable (Prop. 62).

□

The condition is sufficient, not necessary: I has one eigenvalue and is diagonal.

Example 56: diagonalizability depends on the field. $\left[\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right]$ has eigenvalues $2 \pm \sqrt{5}$ (the roots of $x^2 - 4x - 1$), so it is diagonalizable over $\mathbb{R}$; over $\mathbb{Q}$ it has no eigenvalues. $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has eigenvalues $\pm i$, so it is diagonalizable over $\mathbb{C}$ but not over $\mathbb{R}$.

Theorem (Gershgorin disk theorem, LADR 5D)

Let $A = (a_{jk})$ be an $n \times n$ complex matrix. Every eigenvalue of A lies in at least one of the disks

$$D_j = \left\{ z \in \mathbb{C} : |z - a_{jj}| \leq \sum_{k \neq j} |a_{jk}| \right\}, \qquad j = 1, \ldots, n.$$

The idea

Look at the largest coordinate of an eigenvector. In that row the equation $Av = \lambda v$ says that $(\lambda - a_{jj})v_j$ is a combination of the other coordinates, each at most as large as v_j, with the off-diagonal entries of row j as coefficients.

Proof. 1. Let $Av = \lambda v$ with $v \neq 0$, and choose j with $|v_j|$ largest; then $|v_j| > 0$.

2. Row j of $Av = \lambda v$ is $\sum_k a_{jk} v_k = \lambda v_j$, that is,

$$(\lambda - a_{jj}) v_j = \sum_{k \neq j} a_{jk} v_k.$$

3. Take absolute values (triangle inequality, Week 6) and divide by $|v_j|$; since every $|v_k|/|v_j| \leq 1$,

$$|\lambda - a_{jj}| \leq \sum_{k \neq j} |a_{jk}| \frac{|v_k|}{|v_j|} \leq \sum_{k \neq j} |a_{jk}|.$$

□

Instance: for $\left[\begin{smallmatrix}10 & 1\\ 2 & -5\end{smallmatrix}\right]$ the disks are $|z-10|\le 1$ and $|z+5|\le 2$. Neither contains 0, so 0 is not an eigenvalue and the matrix is invertible. (Its eigenvalues are $\frac{5\pm\sqrt{233}}{2}\approx 10.13,\ -5.13$, one in each disk.)

Deciding diagonalizability and diagonalizing a matrix A.

1. Find the eigenvalues λ_i (for a triangular matrix, the diagonal entries).
2. For each, $\dim E(\lambda_i, A) = n - \operatorname{rank}(A-\lambda_i I)$ (T1). A is diagonalizable exactly when these add up to n (Prop. 62).
3. If so, find a basis of each $\operatorname{null}(A-\lambda_i I)$ and put all these vectors as the columns of S. Then $S^{-1}AS = D$ is diagonal, with the eigenvalues in the same order as the columns (T9).

Diagonalizing $\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$. Let $A=\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$. Then $A-3I=\left(\begin{smallmatrix}-1 & 1\\ 1 & -1\end{smallmatrix}\right)$ has null space $\operatorname{span}((1,1))$; $A-I=\left(\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right)$ has null space $\operatorname{span}((1,-1))$. Dimensions $1+1=2$. With $S=\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$: $S^{-1}AS=\left(\begin{smallmatrix}3 & 0\\ 0 & 1\end{smallmatrix}\right)$, and $A^k = S\left(\begin{smallmatrix}3^k & 0\\ 0 & 1\end{smallmatrix}\right)S^{-1}$.

The question answered. There is such a basis exactly when V has a basis of eigenvectors, that is, when the eigenspaces have dimensions adding up to n (Prop. 62). For A it is $(1,1),(1,-1)$; for N there is none, since $\dim E(0,N)=1<2$.

Shortcuts

- **n distinct eigenvalues: diagonalizable.** Reason: Prop. 63. Instance: $\left[\begin{smallmatrix}1 & 4\\ 0 & 2\end{smallmatrix}\right]$ has the eigenvalues $1\neq 2$, so it is diagonalizable.
- **One eigenvalue only: diagonalizable exactly when $T=\lambda I$.** Reason: in an eigenbasis the matrix is diagonal with every entry λ, which is λI, and λI has the matrix λI in every basis. Instance: $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ is not diagonalizable.
- **Dimension of an eigenspace by rank.** $\dim E(\lambda, A) = n - \operatorname{rank}(A-\lambda I)$. Reason: (T1). Instance:

$$A=\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 2\end{pmatrix},\quad \operatorname{rank}(A-I)=2,\quad \text{so } \dim E(1,A)=1.$$

- **If T is diagonalizable, so is every $p(T)$.** Reason: an eigenbasis of T is one of $p(T)$, since $p(T)v=p(\lambda)v$ (T7(b)). Instance: $A=\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$ is diagonalizable, so A^2-A is, with eigenvalues $3^2-3=6$ and $1^2-1=0$.
- **Gershgorin for invertibility.** If no Gershgorin disk contains 0, then A is invertible. Reason: the theorem above and (T2). Instance: the matrix after the theorem.

Traps

- Repeated diagonal entries decide nothing by themselves: I and $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ both have diagonal $1,1$; the first is diagonalizable, the second is not.
- Upper-triangular is not the same as diagonalizable. $\left[\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right]$ is upper-triangular, but it has only the eigenvalue 0 and is not 0, so it is not diagonalizable.
- Diagonalizability depends on the field (Ex. 56).
- T^2 diagonalizable does not make T diagonalizable: $N=\left[\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right]$ has $N^2=0$, which is diagonal, and N is not diagonalizable.

§2.3 Iteration and eigenmodes (Ch. 12.3)

The question. For the three-state Markov chain P below, started at $p_0=(1,0,0)$, what is $P^k p_0$ exactly for every k, and how fast does it approach $(\frac{1}{4},\frac{1}{2},\frac{1}{4})$?

Proposition (Prop. 64: iteration in an eigenbasis)
If $v_1, \dots, v_n$ is an eigenbasis with $Tv_j = \lambda_j v_j$ and $v = a_1 v_1 + \cdots + a_n v_n$, then $T^k v = a_1 \lambda_1^k v_1 + \cdots + a_n \lambda_n^k v_n$ for every $k \geq 0$.

Proof. 1. $T^k v_j = \lambda_j^k v_j$ by induction on k: $T^0 v_j = v_j$, and $T^{k+1} v_j = T(\lambda_j^k v_j) = \lambda_j^k T v_j = \lambda_j^{k+1} v_j$.

2. T^k is linear, so $T^k v = \sum_j a_j T^k v_j = \sum_j a_j \lambda_j^k v_j$.

□

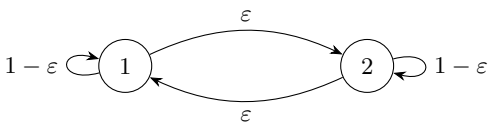
The notes call each $a_j v_j$ an **eigenmode**. Under repeated application of T, a mode with $|\lambda_j| < 1$ shrinks to 0, one with $|\lambda_j| = 1$ keeps its size, and one with $|\lambda_j| > 1$ grows.

Markov chain. Take the stochastic matrix P and the three vectors

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}, \qquad \pi = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix}, \quad u = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad w = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$

1. Multiplying: $P\pi = \pi$, $Pu = \frac{1}{2}u$, $Pw = 0$. So π, u, w is an eigenbasis with eigenvalues $1, \frac{1}{2}, 0$.
2. The start $p_0 = (1, 0, 0)$ is $\pi - \frac{1}{2}u + \frac{1}{4}w$ (check the first coordinate: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$).
3. By Prop. 64, for $k \geq 1$:
$$P^k p_0 = \pi - \frac{1}{2} \cdot \frac{1}{2^k} u + 0 = \pi - 2^{-(k+1)} u.$$
4. So $P^k p_0 \to \pi$, and the distance to π halves at every step. The eigenvalue 1 gives the limit; the next largest eigenvalue, $\frac{1}{2}$, gives the speed.

- **Spectral gap.** $P_\varepsilon = \left(\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\right)$, $0 < \varepsilon < \frac{1}{2}$, has eigenvectors $(1, 1)$ and $(1, -1)$ with eigenvalues 1 and $1 - 2\varepsilon$. So $P_\varepsilon^k(1, -1) = (1 - 2\varepsilon)^k(1, -1)$. When ε is small (moving between the two states is rare), $1 - 2\varepsilon$ is close to 1 and the starting state is forgotten slowly. The number $1 - (1 - 2\varepsilon) = 2\varepsilon$ is the **spectral gap**. A narrow route between two large groups of states, like the small probability ε here, is called a **bottleneck**; a bottleneck gives a small spectral gap and slow convergence.



The chain P_ε: each step stays put with probability $1 - \varepsilon$. The difference between the two probabilities is multiplied by $1 - 2\varepsilon$ at each step, so for small ε it takes about $1/(2\varepsilon)$ steps to shrink noticeably.

- **PageRank** (Week 6). There P is the column-stochastic link matrix, $\alpha \in (0, 1)$ the damping factor, and G the Google matrix, with $\mathbb{R}^n = \operatorname{span}(r) \oplus H$, $Gr = r$, and $Gx = \alpha P x$ for $x \in H$. If $x \in H$ is an eigenvector of P with eigenvalue λ, then $Gx = \alpha\lambda x$. Every eigenvalue of P has modulus at most 1, so these eigenvalues of G have modulus at most α: in an eigenbasis, every mode other than r is multiplied by a factor of modulus at most α per step. (The contraction estimate of Week 6 gave the same convergence without assuming that P is diagonalizable.)

The sequences $(a_0, a_1, a_2, \dots)$ with $a_{k+1} = a_k + a_{k-1}$ form a subspace R, and each one is determined by (a_0, a_1). So $J(a_0, a_1, \dots) = (a_0, a_1)$ is an isomorphism $R \to \mathbb{R}^2$. Under J the shift $S(a_0, a_1, a_2, \dots) = (a_1, a_2, \dots)$ becomes $T(x, y) = (y, x + y)$, so a question about an infinite sequence becomes a question about powers of one operator on $\mathbb{R}^2$.

Fibonacci numbers, LADR 5D Exercise 21. $F_0 = 0$, $F_1 = 1$, $F_{k+1} = F_k + F_{k-1}$. Let $T(x, y) = (y, x + y)$, so $T(F_{k-1}, F_k) = (F_k, F_{k+1})$ and $T^k(0, 1) = (F_k, F_{k+1})$.

1. $T(1, \lambda) = (\lambda, 1 + \lambda)$ equals $\lambda(1, \lambda)$ exactly when $\lambda^2 = \lambda + 1$: $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Two distinct eigenvalues, so $(1, \varphi), (1, \psi)$ is an eigenbasis (Prop. 63).

2. $(0,1) = \frac{1}{\sqrt{5}}\big((1,\varphi) - (1,\psi)\big)$, since $\varphi - \psi = \sqrt{5}$.

3. By Prop. 64, $T^k(0,1) = \frac{1}{\sqrt{5}}\big(\varphi^k(1,\varphi) - \psi^k(1,\psi)\big)$. First coordinate: $F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}}$.

The question answered. $P^k p_0 = \pi - 2^{-(k+1)}u$ for $k \geq 1$ (the Markov example), so the distance to π is halved at every step.

Shortcuts

- **Powers.** Write v in an eigenbasis and multiply each coefficient by λ_j^k, or use $A^k = SD^kS^{-1}$. Reason: Prop. 64 and (T9). Instance: for $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$,

$$A^k = \frac{1}{2}\begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

- **Speed of convergence: the second-largest $|\lambda|$.** Reason: Prop. 64; every mode with $|\lambda_j| < 1$ is multiplied by λ_j at each step. Instance: P_ε, where the factor is $1 - 2\varepsilon$.

Traps

- An eigenvalue of modulus 1 other than 1 prevents convergence: $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has the eigenvalue -1 with eigenvector $(1,-1)$, and $P^k e_1$ alternates between e_1 and e_2.

§2.4 Minimal polynomial (Ch. 13)

The question. For $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $A^2 = 4A - 3I$. Can every power of A be written as a combination of I and A? Which polynomials p have $p(A) = 0$, and what do they tell us about the eigenvalues of A and whether A is diagonalizable?

Definition (Def. 35, 36 / LADR 5.14, 5.22, 5.24)**.** For $p(x) = a_m x^m + \cdots + a_1 x + a_0$, $p(T) = a_m T^m + \cdots + a_1 T + a_0 I$. Let k be the smallest index with $T^k \in \operatorname{span}(I, T, \ldots, T^{k-1})$, and write $T^k = a_{k-1}T^{k-1} + \cdots + a_0 I$. The **minimal polynomial** of T is $m_T(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_0$. (The notes call it p.)

- k exists: $\mathcal{L}(V)$ has dimension n^2 (operators are $n \times n$ matrices), so the $n^2 + 1$ operators $I, T, \ldots, T^{n^2}$ are dependent. Solving a dependence for the highest power with a nonzero coefficient puts that power in the span of the lower ones.
- $I, T, \ldots, T^{k-1}$ are independent, by the same argument: a dependence among them would put some T^j with $j < k$ in the span of lower powers. So the a_i are unique.
- m_T is monic and $m_T(T) = 0$. No nonzero polynomial of degree $< k$ kills T, since it would be a dependence among $I, \ldots, T^{k-1}$. So m_T is the unique monic polynomial of least degree that kills T (two of them would differ by a nonzero polynomial of lower degree that kills T, and dividing it by its leading coefficient would give a monic one).

Example 57. $m_I = x - 1$. For $T = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, I, T are independent (T is not a multiple of I) and

$$T^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = 4T - 3I,$$

so $m_T = x^2 - 4x + 3 = (x-1)(x-3)$.

Proposition (Prop. 65 / LADR 5.29)
For $q \in \mathcal{P}(\mathbb{F})$: $q(T) = 0 \iff m_T \mid q$.

The idea

m_T has the smallest degree among nonzero polynomials that kill T. Polynomials can be divided with remainder, and the remainder has smaller degree than the divisor.

Divide q by m_T. The remainder kills T too, and its degree is too small for it to be nonzero.

Proof. 1. If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$ (T5).

2. If $q(T) = 0$, divide: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$.

3. A nonzero s of degree $< \deg m_T$ cannot kill T. So $s = 0$ and $m_T \mid q$.

□

Proposition (Prop. 66 / LADR 5.27)
The eigenvalues of T are exactly the roots of m_T in $\mathbb{F}$. If $m_T = (x - \lambda)s$, then $s(T) \neq 0$, and every nonzero vector $s(T)w$ is an eigenvector for λ.

The idea

On an eigenvector, a polynomial in T acts as the number $p(\lambda)$ (T7). Apply this to m_T, which is 0 as an operator.

For the converse we need a nonzero vector sent to 0 by $T - \lambda I$. Factor $m_T = (x - \lambda)s$. Then $(T - \lambda I)s(T) = m_T(T) = 0$, so $T - \lambda I$ kills every vector of the form $s(T)w$. Such a vector can be chosen nonzero because s has smaller degree than m_T, so $s(T) \neq 0$.

Proof. 1. If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$.

2. If $m_T(\lambda) = 0$, write $m_T = (x - \lambda)s$ (T6). Since $\deg s < \deg m_T$, $s(T) \neq 0$; pick w with $s(T)w \neq 0$.

3. $(T - \lambda I)\big(s(T)w\big) = m_T(T)w = 0$, so $s(T)w$ is an eigenvector for λ.

The notes argue the second half by contradiction: if $T - \lambda I$ were invertible, multiplying $0 = s(T)(T - \lambda I)$ by $(T - \lambda I)^{-1}$ would give $s(T) = 0$. □

Proposition (Divisors of a product of linear factors)
If a monic polynomial s divides $(x - \lambda_1) \cdots (x - \lambda_r)$, with $\lambda_i \in \mathbb{F}$ not necessarily distinct, then s is the product of some of these factors.

Proof. 1. Induction on r; for $r = 0$ the product is 1 and $s = 1$. Write $st = (x - \lambda_1) \cdots (x - \lambda_r)$.

2. At $x = \lambda_r$ the right side is 0, so $s(\lambda_r) = 0$ or $t(\lambda_r) = 0$. By the factor theorem (T6), $x - \lambda_r$ divides s or t: $s = (x - \lambda_r)s_1$ or $t = (x - \lambda_r)t_1$.

3. Cancel the factor $x - \lambda_r$ from both sides. (If $(x - \lambda)a = (x - \lambda)b$, then $(x - \lambda)(a - b) = 0$, and a nonzero $a - b$ would give a nonzero product, so $a = b$.) This leaves $s_1 t = (x - \lambda_1) \cdots (x - \lambda_{r-1})$ or $st_1 = (x - \lambda_1) \cdots (x - \lambda_{r-1})$.

4. By induction s_1, respectively s, is a product of some of $x - \lambda_1, \ldots, x - \lambda_{r-1}$. So s is a product of some of $x - \lambda_1, \ldots, x - \lambda_r$.

□

Proposition (LADR 5.40, 5.44: upper-triangular matrices and m_T)

(a) If $\mathcal{M}(T)$ is upper-triangular in $v_1, \dots, v_n$ with diagonal $\lambda_1, \dots, \lambda_n$, then $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$. (b) T has an upper-triangular matrix in some basis if and only if $m_T = (x - \lambda_1) \cdots (x - \lambda_m)$ for some $\lambda_i \in \mathbb{F}$ (not necessarily distinct).

The idea

In (a), $T - \lambda_j I$ sends $\operatorname{span}(v_1, \dots, v_j)$ into $\operatorname{span}(v_1, \dots, v_{j-1})$, because the v_j-coefficient of $(T - \lambda_j I)v_j$ is $\lambda_j - \lambda_j = 0$. Applying $T - \lambda_n I$, then $T - \lambda_{n-1} I$, and so on lowers the span by one each time, and after n steps everything is 0. For (b, $\Leftarrow$), copy the proof of Prop. 61, but get the eigenvalue from a root of m_T (Prop. 66) instead of from $\mathbb{C}$.

Proof. (a)

1. Let $U_j = \operatorname{span}(v_1, \dots, v_j)$ and $U_0 = \{0\}$. For $i < j$, $(T - \lambda_j I)v_i \in U_i \subseteq U_{j-1}$ (Prop. 59), and $(T - \lambda_j I)v_j \in U_{j-1}$ since its v_j-coefficient is 0. So $(T - \lambda_j I)U_j \subseteq U_{j-1}$.
2. The factors commute (T5), so apply them to $v \in V = U_n$ in the order $T - \lambda_n I$, then $T - \lambda_{n-1} I$, …, then $T - \lambda_1 I$: the result lies in U_{n-1}, then U_{n-2}, …, then $U_0 = \{0\}$.

(b)

1. ($\Rightarrow$) By (a) the product $(x - \lambda_1) \cdots (x - \lambda_n)$ of the diagonal entries kills T, so m_T divides it (Prop. 65), and m_T is a product of some of its factors (the previous proposition).
2. ($\Leftarrow$) Induction on n; for $n = 1$ every matrix is upper-triangular. m_T has degree ≥ 1 and is a product of linear factors, so it has a root $\lambda \in \mathbb{F}$, which is an eigenvalue (Prop. 66).
3. $U = \operatorname{range}(T - \lambda I)$ is invariant (T5), and $\dim U < n$ because $T - \lambda I$ is not injective (T2). For $u \in U$, $m_T(T|_U)u = m_T(T)u = 0$, so $m_{T|_U} \mid m_T$ (Prop. 65 for $T|_U$), and $m_{T|_U}$ is again a product of linear factors (the previous proposition).
4. By induction $T|_U$ is upper-triangular in a basis $u_1, \dots, u_m$ of U (empty if $U = \{0\}$). Extend it to a basis $u_1, \dots, u_m, v_1, \dots, v_k$ of V. Then $Tv_j = (T - \lambda I)v_j + \lambda v_j \in \operatorname{span}(u_1, \dots, u_m, v_j)$, and Prop. 59 gives an upper-triangular matrix.

□

Instance: over $\mathbb{R}$ the rotation has $m = x^2 + 1$, which is not a product of real linear factors, so it has no upper-triangular matrix over $\mathbb{R}$.

Proposition (Prop. 67 / LADR 5.22)

$\deg m_T \leq n$.

The idea

It is enough to find some polynomial of degree $\leq n$ that kills T, since m_T divides it (Prop. 65).

For one vector $v \neq 0$, the $n + 1$ vectors $v, Tv, \dots, T^n v$ are dependent, so $q(T)v = 0$ for some $q \neq 0$ of degree $\leq n$. This kills v, but not necessarily all of V.

A first attempt is to do this for each basis vector and multiply the q's. The product kills T, but its degree can be about n^2.

Induction on $\dim V$. $U = \operatorname{null} q(T)$ is invariant and contains v. If $U \neq V$, both U and V/U are smaller than V. By induction, a polynomial r of degree $\leq \dim U$ kills T on U, and a polynomial p of degree $\leq \dim V/U$ kills T/U, which means $p(T)$ sends V into U. So $r(T)p(T)$ kills V, and its degree is $\leq \dim U + \dim V/U = n$.

Proof. 1. If $\dim V = 1$, $T = cI$ and $m_T = x - c$. In general, fix $v \neq 0$ and $q \neq 0$ with $\deg q \leq n$ and $q(T)v = 0$. $U = \operatorname{null} q(T)$ is invariant (T5).

2. If $q(T) = 0$, then $m_T \mid q$, so $\deg m_T \leq n$.
3. Otherwise $0 < \dim U < n$. By induction, $r = m_{T|_U}$ has $\deg r \leq \dim U$ and $p = m_{T/U}$ has $\deg p \leq \dim V/U$ (T10).

4. For $v \in V$: $p(T)v + U = p(T/U)(v + U) = 0 + U$, so $p(T)v \in U$, so $r(T)p(T)v = 0$. Hence $m_T \mid rp$, and $\deg(rp) \le \dim U + \dim V/U = n$ (T10).

□

Second proof (LADR 5.22), without quotient spaces. Same plan, with $\operatorname{range} q(T)$ in place of V/U. Induction on n.

1. Fix $v \neq 0$ and let q be the monic polynomial of least degree d with $q(T)v = 0$; $d \ge 1$ since $v \neq 0$.
2. $v, Tv, \ldots, T^{d-1}v$ are independent (a dependence would be a smaller polynomial killing v), and they lie in $\operatorname{null} q(T)$, because $q(T)T^j v = T^j q(T)v = 0$. So $\dim \operatorname{null} q(T) \ge d$.
3. $W = \operatorname{range} q(T)$ is invariant (T5) and $\dim W \le n - d < n$ (T1). By induction some s of degree $\le n - d$ kills $T|_W$.
4. For every u: $q(T)u \in W$, so $s(T)q(T)u = 0$. So sq kills T, and $\deg(sq) \le n$.

Example 58: every degree from 1 to n occurs.

- $m_I = x - 1$ in every dimension.
- The shift N ($Ne_1 = 0$, $Ne_j = e_{j-1}$) has $m_N = x^n$, since $N^{n-1}e_n = e_1 \neq 0$ and $N^n = 0$.
- Degree k, $1 \le k \le n$: let $Ne_1 = 0$, $Ne_j = e_{j-1}$ for $2 \le j \le k$, $Ne_j = 0$ for $j > k$. Then $N^k = 0$ and $N^{k-1}e_k = e_1 \neq 0$, so $m_N = x^k$.

Proposition (Prop. 68 / LADR 5.62)

T is diagonalizable if and only if $m_T = (x - \lambda_1) \cdots (x - \lambda_m)$ for some distinct $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

The idea

($\Rightarrow$) Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues. On an eigenvector for λ_j, the factor $T - \lambda_j I$ gives 0, and the factors commute, so that factor can act first. So $\prod_i (T - \lambda_i I)$ sends every vector of an eigenbasis to 0, and it is the zero map.

($\Leftarrow$) We must write every v as a sum of eigenvectors. Since $m_T(T) = 0$, removing one factor $T - \lambda_j I$ from $m_T(T)$ leaves a map whose outputs are eigenvectors for λ_j (or 0). It remains to choose these pieces so that they add up to v.

Producing the eigenvectors. Factor $T - \lambda_j I$ out of $m_T(T) = 0$:

$$(T - \lambda_j I) \prod_{i \neq j} (T - \lambda_i I) = m_T(T) = 0.$$

So every vector of the form $\prod_{i \neq j}(T - \lambda_i I)w$ is in $E(\lambda_j, T)$. To make the pieces add up to v, scale with the polynomials

$$g_j(x) = \prod_{i \neq j} \frac{x - \lambda_i}{\lambda_j - \lambda_i},$$

which are 1 at λ_j and 0 at the other λ_i, and satisfy $\sum_j g_j = 1$ (step 2 of the proof). Applying both sides to v gives $v = \sum_j g_j(T)v$, and each $g_j(T)v$ is a multiple of a vector of the form $\prod_{i \neq j}(T - \lambda_i I)w$, so it lies in $E(\lambda_j, T)$.

Proof. 1. ($\Rightarrow$) Let $v_1, \ldots, v_n$ be an eigenbasis, $Tv_k = \mu v_k$ with μ one of the λ_i. Letting the factor $T - \mu I$ act first, $\prod_i (T - \lambda_i I)v_k = 0$ for every k (T5). So $m_T \mid \prod_i (x - \lambda_i)$ (Prop. 65). Each λ_i is a root of m_T (Prop. 66), so $m_T = \prod_i (x - \lambda_i)$.

2. ($\Leftarrow$) Define g_j as above. $\sum_j g_j - 1$ has degree $\le m - 1$ and vanishes at the m points $\lambda_1, \ldots, \lambda_m$. A nonzero polynomial of degree $\le m - 1$ has at most $m - 1$ roots (each root splits off a factor, T6), so $\sum_j g_j = 1$.

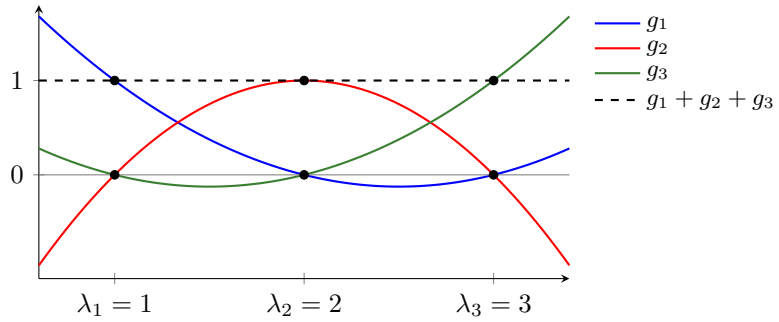
3. For $v \in V$: $v = \sum_j g_j(T)v$, and $(T - \lambda_j I)g_j(T)v = c_j\, m_T(T)v = 0$, where

$$c_j = \prod_{i \neq j} (\lambda_j - \lambda_i)^{-1}.$$

So $g_j(T)v \in E(\lambda_j, T)$.

4. So $V = E(\lambda_1, T) + \cdots + E(\lambda_m, T)$. This sum is direct (T7), so bases of the eigenspaces together form an eigenbasis.

□



g_j is 1 at λ_j and 0 at the other eigenvalues, and $g_1 + g_2 + g_3 = 1$. Applying $1 = \sum g_j$ to T writes $v = \sum g_j(T)v$, and each $g_j(T)v$ lies in $E(\lambda_j, T)$ (Prop. 68).

Second proof of (⇐), the one in the notes. Instead of splitting v into all its eigenvector pieces at once, split off the λ_1-piece and use induction for the rest. Let $g = \prod_{i \geq 2}(x - \lambda_i)$, so $m_T = (x - \lambda_1)g$ and $g(\lambda_1) \neq 0$.

1. Dividing by $x - \lambda_1$ leaves the remainder $g(\lambda_1)$ (T6): $g = (x - \lambda_1)q + g(\lambda_1)$.
2. For $v \in V$ let $u = g(T)v/g(\lambda_1)$. Then $(T - \lambda_1 I)u = m_T(T)v/g(\lambda_1) = 0$, so $u \in \text{null}(T - \lambda_1 I)$, and $v - u = -(T - \lambda_1 I)q(T)v/g(\lambda_1) \in \text{range}(T - \lambda_1 I)$.
3. So $V = \text{null}(T - \lambda_1 I) + \text{range}(T - \lambda_1 I)$. The two dimensions add up to n (T1), so the intersection is 0 (last item of T3) and the sum is direct.
4. $\text{range}(T - \lambda_1 I)$ is invariant, smaller than V (since $\text{null}(T - \lambda_1 I) \neq 0$: λ_1 is an eigenvalue by Prop. 66), and its minimal polynomial divides m_T, because $m_T(T|_U)u = m_T(T)u = 0$ for u in $U = \text{range}(T - \lambda_1 I)$ (then Prop. 65 for $T|_U$). By the proposition on divisors of a product of linear factors, it again is a product of distinct linear factors. By induction $T|_U$ has an eigenbasis; add a basis of $E(\lambda_1, T)$.

The question answered. Every power of A is a combination of I and A: A^k is the remainder of x^k on division by m_A, evaluated at A (Shortcuts, "Any polynomial"). The polynomials with $p(A) = 0$ are the multiples of $m_A = (x-1)(x-3)$ (Prop. 65). Its roots $1, 3$ are the eigenvalues (Prop. 66), and they are distinct, so A is diagonalizable (Prop. 68).

More facts about m_T

Throughout, write $m_T = x^d + c_{d-1}x^{d-1} + \cdots + c_1 x + c_0$.

Proposition (F1: invertibility, HW7 P3)
T is invertible $\iff$ $c_0 \neq 0$. In that case $T^{-1} = -\frac{1}{c_0}\left(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I\right)$, a polynomial in T.

The idea

$m_T(T) = 0$, and every term of $m_T(T)$ except $c_0 I$ contains a factor T.

Move $c_0 I$ to the other side and factor out T. That gives $T \cdot (\text{something}) = -c_0 I$, so $\frac{-1}{c_0}(\text{something})$ is the inverse.

Proof. 1. $c_0 = m_T(0)$. By Prop. 66, $c_0 = 0$ $\iff$ 0 is an eigenvalue $\iff$ T is not injective (T7) $\iff$ T is not invertible (T2).

2. If $c_0 \neq 0$: $m_T(T) = 0$ says $T^d + c_{d-1}T^{d-1} + \cdots + c_1 T = -c_0 I$.
3. Factor T out of the left side: $T\left(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I\right) = -c_0 I$. Divide by $-c_0$.

□

Proposition (F2: degree one, M4)
If $V \neq \{0\}$: $\deg m_T = 1 \iff T = cI$ for some $c \in \mathbb{F}$.

Proof. 1. ($\Rightarrow$) $m_T = x - c$ and $m_T(T) = 0$ say $T - cI = 0$.

2. ($\Leftarrow$) $x - c$ kills cI. No polynomial of degree 0 does: a nonzero constant a gives $aI \neq 0$ since $V \neq \{0\}$. So $m_{cI} = x - c$.

□

Proposition (F3: restrictions and quotients, Ex. 60)
Let U be T-invariant.

(a) $m_{T|_U} \mid m_T$ and $m_{T/U} \mid m_T$.

(b) A monic divisor of a product $(x - \lambda_1) \cdots (x - \lambda_m)$ of distinct linear factors is the product of some of those factors.

(c) So if T is diagonalizable, so are $T|_U$ and T/U.

Proof. (a)

1. For $u \in U$: $m_T(T|_U)u = m_T(T)u = 0$. So m_T kills $T|_U$, and $m_{T|_U} \mid m_T$ by Prop. 65 applied to $T|_U$.
2. For $v + U \in V/U$: $m_T(T/U)(v + U) = m_T(T)v + U = 0 + U$ (T10). So $m_{T/U} \mid m_T$.

(b) Let s be the monic divisor, $st = \prod_{i=1}^{m}(x - \lambda_i)$.

1. Each λ_i makes the product 0, so it is a root of s or of t. Let S be the set of λ_i that are roots of s, and R the remaining λ_i (roots of t).
2. Distinct roots split off one factor each (T6), so $\prod_{\lambda \in S}(x - \lambda)$ divides s and $\prod_{\lambda \in R}(x - \lambda)$ divides t. Hence $\deg s \geq |S|$ and $\deg t \geq |R|$.
3. But $\deg s + \deg t = m = |S| + |R|$, so both are equalities. A monic s divisible by a monic polynomial of the same degree equals it: $s = \prod_{\lambda \in S}(x - \lambda)$.

(c) m_T is a product of distinct linear factors (Prop. 68). By (a) and (b), so are $m_{T|_U}$ and $m_{T/U}$. Apply Prop. 68 again. □

Definition (Least common multiple)**.** For nonzero polynomials $p_1, \ldots, p_r$, a **common multiple** is a polynomial divisible by every p_j. The **least common multiple** $\operatorname{lcm}(p_1, \ldots, p_r)$ is the monic common multiple of least degree.

Proposition (The lcm divides every common multiple)
$L = \operatorname{lcm}(p_1, \ldots, p_r)$ exists, is unique, and divides every common multiple. If $p_j = \prod_\lambda (x - \lambda)^{e_{j\lambda}}$ for all j, then $L = \prod_\lambda (x - \lambda)^{\max_j e_{j\lambda}}$.

Proof. 1. The product $p_1 \cdots p_r$ is a common multiple, so a monic common multiple L of least degree exists.

2. Let q be any common multiple and divide: $q = Lh + s$ with $s = 0$ or $\deg s < \deg L$.
3. Each p_j divides q and Lh, hence divides $s = q - Lh$. A nonzero s, divided by its leading coefficient, would be a monic common multiple of smaller degree than L, which is impossible. So $s = 0$ and $L \mid q$. Two such L divide each other and are monic, so they are equal.
4. For the formula: $\prod_\lambda (x - \lambda)^{\max_j e_{j\lambda}}$ is a common multiple, so L divides it, and L is a product of some of its factors (divisors of a product of linear factors). L is divisible by each p_j, hence by $(x - \lambda)^{e_{j\lambda}}$ for every j; so the exponent of $x - \lambda$ in L is at least $\max_j e_{j\lambda}$. Hence equality.

□

Proposition (F4: direct sums, block diagonal matrices)

If $V = U_1 \oplus \cdots \oplus U_r$ with every U_i invariant, then $m_T = \operatorname{lcm}\big(m_{T|_{U_1}}, \ldots, m_{T|_{U_r}}\big)$. For a block diagonal matrix, m is the lcm of the minimal polynomials of the blocks.

Proof. 1. Write $v = u_1 + \cdots + u_r$ with $u_i \in U_i$. Then $q(T)v = \sum_i q(T|_{U_i})u_i$, so $q(T) = 0 \iff q(T|_{U_i}) = 0$ for every i.

2. By Prop. 65, that holds $\iff$ every $m_{T|_{U_i}}$ divides q.

3. So m_T is the monic polynomial of least degree divisible by all the $m_{T|_{U_i}}$: their lcm. The blocks of the matrix are the matrices of the $T|_{U_i}$ (T8).

□

Proposition (F5: similarity, transpose, dual)

Similar matrices have the same minimal polynomial. Also $m_{A^\intercal} = m_A$ and $m_{T'} = m_T$.

Proof. 1. $(S^{-1}AS)^k = S^{-1}A^kS$ (the inner factors SS^{-1} cancel), so $p(S^{-1}AS) = S^{-1}p(A)S$. This is 0 exactly when $p(A) = 0$. So the same polynomials kill both, and the minimal polynomials agree.

2. $(A^\intercal)^k = (A^k)^\intercal$, so $p(A^\intercal) = p(A)^\intercal$, which is 0 exactly when $p(A) = 0$.

3. In dual bases, $\mathcal{M}(T') = \mathcal{M}(T)^\intercal$; apply step 2.

□

Proposition (F6: shifts, scalings, inverses, HW7 P4, M5)

With $d = \deg m_T$:

(a) $m_{T-\lambda I}(x) = m_T(x + \lambda)$;

(b) $m_{cT}(x) = c^d\, m_T(x/c)$ for $c \neq 0$;

(c) if T is invertible, $m_{T^{-1}}(x) = \frac{1}{c_0}x^d m_T(1/x)$, i.e. reverse the list of coefficients of m_T and divide by c_0.

The idea

A polynomial in the new operator is a polynomial in T after a substitution: $q(T - \lambda I)$ is $r(T)$ with $r(x) = q(x - \lambda)$.

The substitution can be undone and does not change degrees. So it matches the polynomials that kill the new operator with the polynomials that kill T, and the one of least degree goes to the one of least degree.

Proof. (a)

1. For any q, $q(T - \lambda I) = r(T)$ with $r(x) = q(x - \lambda)$.

2. So q kills $T - \lambda I \iff m_T(x) \mid q(x - \lambda)$ (Prop. 65) $\iff m_T(x + \lambda) \mid q(x)$ (replace x by $x + \lambda$ throughout).

3. $m_T(x + \lambda)$ is monic, so it is $m_{T-\lambda I}$.

(b)

1. For any q, $q(cT) = r(T)$ with $r(x) = q(cx)$.

2. So q kills $cT \iff m_T(x) \mid q(cx)$ (Prop. 65) $\iff m_T(x/c) \mid q(x)$ (replace x by x/c throughout; this substitution can be undone and keeps degrees).

3. $m_T(x/c)$ has leading coefficient c^{-d}, so $c^d m_T(x/c)$ is monic, divides the same polynomials, and is m_{cT}.

(c)

1. Let
$$r(x) = \frac{1}{c_0}x^d m_T(1/x) = \frac{1}{c_0}\big(c_0x^d + c_1x^{d-1} + \cdots + c_{d-1}x + 1\big).$$
It is monic of degree d ($c_0 \neq 0$ by F1).

2. Substitute T^{-1} into r and factor out T^{-d}:

$$r(T^{-1}) = \frac{1}{c_0}\left(c_0 T^{-d} + c_1 T^{-(d-1)} + \cdots + I\right) = \frac{1}{c_0} T^{-d} m_T(T) = 0.$$

   So $m_{T^{-1}} \mid r$ and $\deg m_{T^{-1}} \leq d$.

3. The same argument with T^{-1} in place of T (its inverse is T) gives $d \leq \deg m_{T^{-1}}$.

4. So $m_{T^{-1}}$ divides r and has the same degree; both are monic, so they are equal.

□

Proposition (F7: one vector at a time)

For $v \in V$, let p_v be the monic polynomial of least degree with $p_v(T)v = 0$.

(a) $q(T)v = 0 \iff p_v \mid q$. In particular $p_v \mid m_T$.

(b) For a basis $v_1, \ldots, v_n$: $m_T = \operatorname{lcm}(p_{v_1}, \ldots, p_{v_n})$.

(c) If $v, Tv, \ldots, T^{n-1}v$ is a basis (v is a **cyclic vector**), then $m_T = p_v$, which has degree n.

The idea

Polynomials that kill one vector behave like polynomials that kill T: the division argument of Prop. 65 works the same way. An operator that is 0 on a basis is 0.

Killing T means killing every basis vector, i.e. being divisible by every p_{v_i}. The least such polynomial is the lcm.

Proof. 1. (a) Divide: $q = p_v s + r$ with $\deg r < \deg p_v$ or $r = 0$. If $q(T)v = 0$, then $r(T)v = q(T)v - s(T)p_v(T)v = 0$, so $r = 0$ by minimality of p_v. Since $m_T(T)v = 0$, $p_v \mid m_T$.

2. (b) Let L be the lcm. Each $p_{v_i} \mid L$, so $L(T)v_i = 0$ by (a); $L(T)$ is 0 on a basis, so $L(T) = 0$ and $m_T \mid L$.

3. Each $p_{v_i} \mid m_T$ by (a), so $L \mid m_T$. Both are monic, so $L = m_T$.

4. (c) $v, \ldots, T^{n-1}v$ are independent, so no nonzero polynomial of degree $< n$ kills v: $\deg p_v \geq n$. Also $p_v \mid m_T$ and $\deg m_T \leq n$ (Prop. 67). So $m_T = p_v$.

□

Proposition (F8: real versus complex)

For a real matrix A, m_A is the same over $\mathbb{R}$ and over $\mathbb{C}$. So A is diagonalizable over $\mathbb{C} \iff m_A$ has distinct roots, and diagonalizable over $\mathbb{R} \iff$ in addition every root is real.

The idea

Def. 36 finds m_A by solving linear equations whose coefficients are entries of powers of A, all real. Solving linear equations never leaves the real numbers.

Proof. 1. For each k, "$A^k = a_{k-1}A^{k-1} + \cdots + a_0 I$" is a system of n^2 linear equations in $a_0, \ldots, a_{k-1}$ with real coefficients.

2. Row-reduce the real system with real arithmetic. Whether it is solvable is read off the reduced form (no row $0 = c$ with $c \neq 0$), so the answer is the same over $\mathbb{R}$ and $\mathbb{C}$, and when it is solvable, the reduced form gives a real solution.

3. So the first solvable k is the same over $\mathbb{R}$ and $\mathbb{C}$, and the solution is unique (Def. 36), hence real. So m_A is the same.

4. The consequences are Prop. 68 over each field: over $\mathbb{R}$ the linear factors must have real roots.

□

Proposition (F9: projections onto eigenspaces, HW7 P7)

If $m_T = \prod_{i=1}^{m}(x-\lambda_i)$ with distinct roots, then $g_j(T)$, with g_j as in Prop. 68, is the projection onto $E(\lambda_j, T)$ along $\bigoplus_{i\neq j} E(\lambda_i, T)$.

Proof. 1. Since m_T has distinct roots, $V = \bigoplus_i E(\lambda_i, T)$ (Prop. 68); write $v = v_1 + \cdots + v_m$ with $v_i \in E(\lambda_i, T)$.

2. On $E(\lambda_i, T)$, $g_j(T)$ multiplies by $g_j(\lambda_i)$ (T7(b)), which is 1 if $i = j$ and 0 otherwise.

3. So $g_j(T)v = v_j$: this map fixes $E(\lambda_j, T)$ and sends $\bigoplus_{i\neq j} E(\lambda_i, T)$ to 0, which is the projection. These are the polynomials drawn in the figure after Prop. 68.

□

The **greatest common divisor** $\gcd(p, q)$ of nonzero polynomials is the monic common divisor of largest degree; the Euclidean algorithm below computes it.

Proposition (F10: repeated roots, Ex. 59)

Let p be a nonzero polynomial.

(a) λ is a repeated root of $p \iff p(\lambda) = p'(\lambda) = 0$.

(b) The common roots of p and p' are the roots of $\gcd(p, p')$, which the Euclidean algorithm finds without factoring p.

(c) Over $\mathbb{C}$, if $T^k = I$ for some $k \geq 1$, then T is diagonalizable.

The idea

From calculus: at a double root the graph touches the axis, so the value and the slope are both 0.

The product rule turns this into algebra: differentiating $(x-\lambda)^2 q$ leaves a factor $x - \lambda$.

Proof. (a)

1. If $p = (x-\lambda)^2 q$, then $p' = 2(x-\lambda)q + (x-\lambda)^2 q'$, which is 0 at λ.

2. Conversely, if $p(\lambda) = 0$, write $p = (x-\lambda)q$ (T6). Then $p' = q + (x-\lambda)q'$, so $p'(\lambda) = q(\lambda)$. If this is 0, then $(x-\lambda) \mid q$, so $(x-\lambda)^2 \mid p$.

(b)

1. Divide p by p', then p' by the remainder, and so on, until the remainder is 0; let g be the last nonzero remainder, made monic.

2. Each remainder is (previous dividend) − (quotient)(previous divisor). So a common root of p, p' is a root of every remainder, in particular of g.

3. Conversely, g divides the last divisor, and working back up the divisions, it divides every earlier remainder, p' and p. So the roots of g are exactly the common roots of p and p', and $g = \gcd(p, p')$.

Euclidean algorithm. For $p = x^3 - 3x + 2$ and $p' = 3x^2 - 3$:

$$p = \tfrac{x}{3}\,p' + (-2x+2), \qquad p' = -\tfrac{3}{2}(x+1)\cdot(-2x+2) + 0.$$

So $g = x - 1$, and 1 is a double root: $p = (x-1)^2(x+2)$.

(c)

1. $T^k = I$ means $x^k - 1$ kills T, so $m_T \mid x^k - 1$.

2. $(x^k-1)' = kx^{k-1}$ has only the root 0, which is not a root of $x^k - 1$. By (a), $x^k - 1$ has no repeated root, so it is a product of k distinct linear factors.

3. m_T is a monic divisor of it, so it is a product of distinct linear factors (F3(b)). Prop. 68.

□

Proposition (F11: odd dimension, LADR 5B)

Every operator on an odd-dimensional real vector space has an eigenvalue.

The idea

A real eigenvalue is the same as a real root of m_T (Prop. 66).

Over $\mathbb{R}$ there may be none: the rotation of $\mathbb{R}^2$ has $m = x^2 + 1$. But $\mathbb{R}^2$ has even dimension.

The proof shows that "no real root" forces the dimension to be even, in three steps. (A) A real polynomial with no real root is a product of quadratics $x^2 + bx + c$ with $b^2 < 4c$. (B) For such a quadratic p, $\dim \operatorname{null} p(T)$ is even: each nonzero u in it comes with Tu, and $\operatorname{span}(u, Tu)$ is 2-dimensional. (C) Split off $\operatorname{null} p(T)$ and use induction on the dimension.

Step A: real polynomials. A real polynomial is a product of real linear factors and real quadratics $x^2 + bx + c$ with $b^2 < 4c$.

1. For real p, $p(\bar{z}) = \overline{p(z)}$, so if z is a non-real root, so is $\bar{z}$.
2. The product of the two factors has real coefficients:

$$q = (x - z)(x - \bar{z}) = x^2 - 2(\operatorname{Re} z)x + |z|^2,$$

   and $b^2 - 4c = -4(\operatorname{Im} z)^2 < 0$.
3. Divide over $\mathbb{R}$: $p = qs + r$ with $\deg r < 2$. Then $r(z) = r(\bar{z}) = 0$, and a polynomial of degree < 2 with two roots is 0. So $p = qs$ with s real. Real roots split off real linear factors (T6). Repeat on s (induction on the degree).

Step B: if $b^2 < 4c$ and $p = x^2 + bx + c$, then $\dim \operatorname{null} p(T)$ is even. Claim, for every finite-dimensional real space X and every $S \in \mathcal{L}(X)$: $\dim \operatorname{null} p(S)$ is even. Induction on $\dim \operatorname{null} p(S)$. Let $U = \operatorname{null} p(T)$, which is invariant (T5).

1. $T|_U$ has no eigenvalue: $Tu = \lambda u$ with $u \in U$, $u \neq 0$, gives $0 = p(T)u = p(\lambda)u$, but

$$p(\lambda) = (\lambda + \frac{b}{2})^2 + c - \frac{b^2}{4} > 0.$$

2. If $U = 0$, its dimension 0 is even. Otherwise pick $u \in U$, $u \neq 0$, and let $W = \operatorname{span}(u, Tu)$. It has dimension 2, since u is not an eigenvector. It is invariant, since $T(Tu) = -bTu - cu$.
3. The quotient operator $S = T|_U/W$ on U/W satisfies $p(S) = 0$ on U/W (T10), so $\operatorname{null} p(S) = U/W$, which has dimension $\dim U - 2 < \dim U$. By the induction hypothesis (the claim for $X = U/W$) this is even, so $\dim U$ is even.

Step C: proof of F11. Induction on the odd number $n = \dim V$; $n = 1$ is clear.

1. Suppose T has no eigenvalue. Then m_T has no real root (Prop. 66), so by Step A, $m_T = ps$ with $p = x^2 + bx + c$, $b^2 < 4c$.
2. $s(T) \neq 0$ (smaller degree than m_T), so some w has $s(T)w \neq 0$, and $p(T)\big(s(T)w\big) = 0$. So $\operatorname{null} p(T) \neq 0$, and its dimension is even by Step B.
3. $W = \operatorname{range} p(T)$ is invariant, and $\dim W = n - \dim \operatorname{null} p(T)$ is odd and less than n (T1). $T|_W$ has no eigenvalue, because an eigenvector of $T|_W$ is one of T. This contradicts the induction hypothesis.

Computing m_T

Over $\mathbb{C}$, or whenever m_T splits into linear factors, $m_T = \prod_i (x - \lambda_i)^{k_i}$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues (Prop. 66), every $k_i \geq 1$, and $\sum_i k_i \leq n$ (Prop. 67). So the work is finding the eigenvalues and then the exponents. Use whichever method below needs the least computation.

Method 1: from the definition. Compute $I, A, A^2, \dots$ and stop at the first power that is a combination of the earlier ones. Best for 2×2 matrices or when powers are easy.

Method 1. $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$ is not a multiple of I, so I, A are independent. $A^2 = \left(\begin{smallmatrix} 4 & 5 \\ 0 & 9 \end{smallmatrix}\right)$. Solve $A^2 = aA + bI$ entry by entry: entry $(1,2)$ gives $a = 5$; entry $(1,1)$ gives $4 = 2a + b$, so $b = -6$; entry $(2,2)$ checks: $9 = 3a + b$. So $m_A = x^2 - 5x + 6 = (x-2)(x-3)$.

Method 2: eigenvalues, then test candidates. Find the eigenvalues (for a triangular matrix, the diagonal entries, Prop. 60). The candidates are $\prod_i (x - \lambda_i)^{k_i}$ with every $k_i \geq 1$ and $\sum k_i \leq n$. Test them in increasing degree; the first one that kills A is m_A. Reason: m_A is one of the candidates, and any candidate q with $q(A) = 0$ is a multiple of m_A (Prop. 65), so $\deg q > \deg m_A$ unless $q = m_A$.

Method 2, two examples. Take

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

which has eigenvalues $3, 3, 1$. Candidates: degree 2: $(x-3)(x-1)$; degree 3: $(x-3)^2(x-1)$ and $(x-3)(x-1)^2$.

$$(A - 3I)(A - I) = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0,$$

and $(A - 3I)^2(A - I) = 0$. So $m_A = (x-3)^2(x-1)$, and A is not diagonalizable.

Next take

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

which has eigenvalues $1, 1, 2$, and $(B - I)(B - 2I) = 0$. So $m_B = (x-1)(x-2)$, and B is diagonalizable although it is not diagonal.

Method 3: blocks. For a block diagonal matrix, m is the lcm of the blocks' minimal polynomials (F4). For a **block upper-triangular** matrix $A = \left[\begin{smallmatrix} B & * \\ 0 & C \end{smallmatrix}\right]$ (a matrix whose lower-left block, below the first k rows and left of the last $n - k$ columns, is 0): $m_B \mid m_A$, $m_C \mid m_A$ and $m_A \mid m_B m_C$. So if m_B and m_C have no common root, $m_A = m_B m_C$.

Why.

1. The first block of basis vectors spans an invariant subspace U with $B = \mathcal{M}(T|_U)$. C is the matrix of T/U in the cosets of the remaining basis vectors.
2. $m_B \mid m_A$ and $m_C \mid m_A$ by F3(a). $m_A \mid m_B m_C$ by step 4 of the proof of Prop. 67: $p = m_{T/U}$ sends V into U and $r = m_{T|_U}$ is 0 on U, so $r(T)p(T) = 0$.
3. "No common root" means no common root in $\mathbb{C}$. Over $\mathbb{C}$, $m_A = \prod (x - \lambda)^{e_\lambda}$, and each monic divisor of it is $\prod (x - \lambda)^{b_\lambda}$ with $b_\lambda \leq e_\lambda$ (its roots are roots of m_A; divide out one factor at a time, T6).
4. So m_B and m_C are such products over disjoint sets of λ's, and their product divides m_A.

Method 3, two examples. Take

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

The first block R has $R^2 = -I$ and is not a multiple of I, so $m_R = x^2 + 1$; the block $2I$ has $m = x - 2$. So $m_A = (x-2)(x^2+1)$. Over $\mathbb{C}$ the roots $2, \pm i$ are distinct, so A is diagonalizable over $\mathbb{C}$; over $\mathbb{R}$ it is not, since $\pm i \notin \mathbb{R}$ (F8).

The matrix A of Method 2 is $\left(\begin{smallmatrix} B & * \\ 0 & 1 \end{smallmatrix}\right)$ with $B = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, $m_B = (x-3)^2$. The roots 3 and 1 differ, so $m_A = (x-3)^2(x-1)$ with no matrix multiplication.

Method 4: follow one vector (F7). Compute $v, Av, A^2v, \ldots$ until the first dependence; it gives p_v, which divides m_A. If $\deg p_v = n$, then $m_A = p_v$. Otherwise repeat with more basis vectors and take the lcm.

Method 4, two examples. Take

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}.$$

$Ae_1 = e_2$, $Ae_2 = e_3$, so e_1, Ae_1, A^2e_1 is a basis. $A^3e_1 = Ae_3 = (2,-1,2) = 2e_1 - Ae_1 + 2A^2e_1$. So $m_A = x^3 - 2x^2 + x - 2 = (x-2)(x^2+1)$.

Next take

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

$Ae_1 = (2,1,0)$, $A^2e_1 = (4,4,0) = 4Ae_1 - 4e_1$, so $p_{e_1} = (x-2)^2$. $Ae_3 = 3e_3$, so $p_{e_3} = x-3$. The lcm $(x-2)^2(x-3)$ has degree $3 = n$, so it is m_A; there is no need to compute p_{e_2}.

The first matrix is a **companion matrix**: if $Ae_j = e_{j+1}$ for $j < n$ and $Ae_n = a_0e_1 + \cdots + a_{n-1}e_n$, then e_1 is cyclic and $A^ne_1 = Ae_n$, so $m_A = x^n - a_{n-1}x^{n-1} - \cdots - a_0$. So every monic polynomial is the minimal polynomial of some matrix.

Method 5: spot a relation. If you notice $p(A) = 0$, then $m_A \mid p$; check which divisors work.

- Rank one, $A = uv^{\mathsf{T}}$ (all columns multiples of u):

  $$A^2 = u(v^{\mathsf{T}}u)v^{\mathsf{T}} = (\operatorname{tr} A)A,$$

  where $\operatorname{tr} A$ is the sum of the diagonal entries, which equals $v^{\mathsf{T}}u$. For $n \geq 2$, A is not a multiple of I, so $m_A = x(x - \operatorname{tr} A)$ if $\operatorname{tr} A \neq 0$ (diagonalizable) and $m_A = x^2$ if $\operatorname{tr} A = 0$ (not diagonalizable). Example: $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)^2 = 2\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, so $m = x(x-2)$.
- $A^2 = A$: m_A is x, $x-1$ or $x(x-1)$. $A^2 = I$: m_A is $x-1$, $x+1$ or $(x-1)(x+1)$. Both cases are diagonalizable. $A^k = 0$: $m_A = x^j$ for the least j with $A^j = 0$.
- A permutation matrix P satisfies $P^k = I$ for some $k \geq 1$. There are only finitely many permutation matrices of size n, so two powers agree: $P^a = P^b$ with $a < b$. P is invertible, so multiplying by P^{-a} gives $P^{b-a} = I$. Hence P is diagonalizable over $\mathbb{C}$ (F10(c)).

Method 6: the 2×2 formula. For 2×2 $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, with $\operatorname{tr} A = a+d$ and $\det A = ad-bc$: $m_A = x^2 - (\operatorname{tr} A)x + \det A$, unless $A = \mu I$ for a scalar μ (then $m_A = x - \mu$).

- Why: for $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $A^2 - (a+d)A = \left(\begin{smallmatrix} bc-ad & 0 \\ 0 & bc-ad \end{smallmatrix}\right)$, so $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq \mu I$, $\deg m_A = 2$ (F2) and m_A divides this monic quadratic, so they are equal.

Method 6. For $A = \left[\begin{smallmatrix} 1 & 4 \\ -1 & -3 \end{smallmatrix}\right]$: $\operatorname{tr} A = -2$ and $\det A = 1$, so $m_A = (x+1)^2$; $A \neq -I$, so A is not diagonalizable.

Method 7: operators on function spaces. Work with the operator directly, or take its matrix in a basis.

Method 7. D on $V = \operatorname{span}(e^x, xe^x)$: $(D-I)e^x = 0$, $(D-I)(xe^x) = e^x \neq 0$, and $(D-I)^2(xe^x) = 0$. So $(D-I)^2 = 0$ and $D \neq I$: $m_D = (x-1)^2$, and D is not diagonalizable.

Shortcuts

- **No computation needed.** n distinct eigenvalues (for example a triangular matrix with distinct diagonal entries) give $m = \prod(x - \lambda_i)$, so T is diagonalizable. One eigenvalue λ and $A \neq \lambda I$ give a non-diagonalizable A. Reason: every eigenvalue is a root of m (Prop. 66) and $\deg m \leq n$ (Prop. 67); Prop. 68. Instance:

  $$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix} \text{ has } m = (x-1)(x-4)(x-6).$$

- **Prune candidates.** Every eigenvalue is a root of m (Prop. 66), so skip any candidate that omits one. Instance: in Method 2, $(x-3)^2$ is not a candidate for A, since 1 is an eigenvalue.
- **Testing a candidate.** $q(A)=0$ needs every entry to be 0, so as soon as one computed entry is nonzero, the candidate fails. Instance: in Method 2, the $(1,2)$ entry of $(A-3I)(A-I)$ is 2.
- **Inverse** (F1). Instance: $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$ has $A^2-5A+6I=0$, so $A(5I-A)=6I$ and $A^{-1}=\frac{1}{6}(5I-A)$.
- **High powers.** Divide: $x^k=m(x)q(x)+r(x)$ with $\deg r<\deg m$; then $A^k=r(A)$. Find r by putting each root λ of m into this identity: $r(\lambda)=\lambda^k$. For a double root, use the derivative as well: $r'(\lambda)=k\lambda^{k-1}$. (Why: $x^k-r(x)=m(x)q(x)$ is divisible by $(x-\lambda)^2$, so by F10(a) its value and its derivative are 0 at λ: $\lambda^k=r(\lambda)$ and $k\lambda^{k-1}=r'(\lambda)$.) Instances:
  - $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$, $r=ax+b$: $2a+b=2^k$, $3a+b=3^k$, so $A^k=(3^k-2^k)A+(3\cdot 2^k-2\cdot 3^k)I$.
  - $m=(x+1)^2$ (as for $\left[\begin{smallmatrix}1&4\\-1&-3\end{smallmatrix}\right]$): $r(-1)=(-1)^k$, $r'(-1)=k(-1)^{k-1}$, so $A^k=(-1)^kI+k(-1)^{k-1}(A+I)$.
- **Any polynomial** $p(A)$ equals $r(A)$, where r is the remainder of p divided by m_A. So every polynomial in A is a combination of $I,A,\dots,A^{d-1}$. Reason: $p=m_Aq+r$ and $m_A(A)=0$. Instance: for $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$, $x^3=(x^2-5x+6)(x+5)+19x-30$, so $A^3=19A-30I$.
- **Eigenvectors without solving** (Prop. 66). If $m=(x-\lambda)s$, the nonzero columns of $s(A)$ are eigenvectors for λ; if A is diagonalizable, they span $E(\lambda)$ (on $E(\mu)$, $s(A)$ is the number $s(\mu)$, which is 0 for $\mu\neq\lambda$ and nonzero for $\mu=\lambda$). Instance: for $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$ and $\lambda=3$,

$$s(A)=A-2I=\begin{pmatrix}0&1\\0&1\end{pmatrix},$$

  so $(1,1)$ is an eigenvector for 3.
- **Diagonalizable without finding eigenvalues.** One polynomial that kills T and has distinct roots in $\mathbb{F}$ is enough, since m_T divides it. Reason: the proposition on divisors of a product of linear factors, and Prop. 68. Condition: the roots must be distinct and lie in $\mathbb{F}$. Instance: $T^2=T$, $T^2=I$, $T^3=T$ (over $\mathbb{R}$), $T^k=I$ (over $\mathbb{C}$); the cyclic permutation matrix P of M1(b) has $P^3=I$, so it is diagonalizable over $\mathbb{C}$ (over $\mathbb{R}$, x^3-1 has non-real roots, so this test says nothing).
- **Diagonalizable $\iff$ $V=\operatorname{null}(T-\lambda I)\oplus\operatorname{range}(T-\lambda I)$ for every eigenvalue** λ (over $\mathbb{C}$). Reason: ($\Rightarrow$) in an eigenbasis, $\operatorname{null}(T-\lambda I)$ is spanned by the eigenvectors for λ and $\operatorname{range}(T-\lambda I)$ by the others; ($\Leftarrow$) the second proof of Prop. 68 builds the eigenbasis from this splitting. Instance: for $N=\left[\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right]$, $\operatorname{null}N=\operatorname{range}N=\operatorname{span}(e_1)$, so the sum is not direct.
- **Not similar.** Different minimal polynomials prove it (F5). Equal minimal polynomials do not prove similarity. Instance: $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$ and I ($m=(x-1)^2$ versus $x-1$) are not similar. $\operatorname{diag}(1,1,2)$ and $\operatorname{diag}(1,2,2)$ both have $m=(x-1)(x-2)$ but are not similar: they have different $\dim E(1)$.
- **Lower bound.** If $v,Av,\dots,A^{j-1}v$ are independent, then $\deg m_A\geq j$ (F7). Instance: Method 4, first matrix: e_1,Ae_1,A^2e_1 are independent, so $\deg m_A=3$.

Traps

- Only roots in $\mathbb{F}$ are eigenvalues: over $\mathbb{R}$, $m=x^2+1$ gives no eigenvalues.
- A polynomial that kills T need not be m_T: $T^2=T$ allows $m_T=x$, $x-1$ or x^2-x. Check the divisors.
- The exponent of $x-\lambda$ in m_T is not $\dim E(\lambda,T)$, in either direction. For I_2, $\dim E(1)=2$ but $m=x-1$ has exponent 1. For $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$, $\dim E(1)=1$ but $m=(x-1)^2$ has exponent 2.
- For a triangular matrix, m depends on the entries above the diagonal: I_2 and $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$ have the same diagonal, but $m=x-1$ and $m=(x-1)^2$.
- $p(T)v=0$ for one $v\neq 0$ only gives $p_v\mid p$, not $m_T\mid p$. And v need not be an eigenvector even when p has distinct roots: $T=\operatorname{diag}(1,2)$, $v=(1,1)$, $p=(x-1)(x-2)$. What does follow, if p splits into linear factors over $\mathbb{F}$ (always true over $\mathbb{C}$), is that some root of p is an eigenvalue (T4, T6). Over $\mathbb{R}$ not even that: $T=\left[\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right]$, $p=x^2+1$, $v=e_1$.

- Over $\mathbb{R}$, $T^k = I$ does not imply diagonalizable: $R = \left[\begin{smallmatrix} 1 & -2 \\ 1 & -1 \end{smallmatrix}\right]$ has $R^4 = I$ but $m_R = x^2 + 1$.
- $m_{T^{-1}}(x)$ is not $m_T(1/x)$, which is not a polynomial. For $T = \operatorname{diag}(1, 2)$, $m_T(1/x) = (1/x - 1)(1/x - 2)$, while $m_{T^{-1}} = (x - 1)(x - \frac{1}{2})$: reverse the coefficients and divide by c_0.
- $\deg m_T = n$ says nothing about diagonalizability: $\operatorname{diag}(1, 2)$ (diagonalizable) and $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ (not diagonalizable) both have $\deg m = 2$.

§3 Problem set

HW7 P1. Return to the matrices from Homework 6,

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(a) Find the minimal polynomial of A over $\mathbb{R}$ and over $\mathbb{C}$. (b) Find the minimal polynomial of B over $\mathbb{R}$. (c) Use the minimal-polynomial criterion for diagonalizability to recover the conclusions of Homework 6 about whether these operators are diagonalizable over the indicated fields.

Hint: $A^2 = -I$ and $(B - I)^2 = 0$; neither matrix is a multiple of I, so $\deg m = 2$ (F2); then Prop. 68 and F8.

HW7 P2. Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation. (a) Find the minimal polynomial of D. (b) Decide whether D is diagonalizable.

Hint: $D^4 = 0$ but $D^3 x^3 = 6$ (Ex. 54); then Prop. 68.

HW7 P3. Let T be an invertible operator on a finite-dimensional vector space, and let $m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$ be its minimal polynomial. (a) Prove that $a_0 \neq 0$. (b) Prove that T^{-1} is a polynomial in T. (c) Suppose now that V is a complex vector space and that

$$m_T(x) = x^4 + x^3 - x^2 - 5x + 4.$$

Express T^{-1} explicitly as a polynomial in T, find the eigenvalues of T, and decide whether T is diagonalizable.

Hint: (a), (b): F1. (c): $m_T(1) = m_T'(1) = 0$ (F10(a)), so divide by $(x - 1)^2$; then Props. 66 and 68.

HW7 P4 (optional). Let m_T be the minimal polynomial of an operator T. (a) Prove that the minimal polynomial of $T - \lambda I$ is $m_T(x + \lambda)$. (b) If $c \neq 0$ and $d = \deg m_T$, prove that the minimal polynomial of cT is $c^d m_T(x/c)$.

Hint: F6(a), (b): a substitution $x \mapsto x + \lambda$ or $x \mapsto x/c$ matches the polynomials that are 0 at the two operators (Prop. 65).

HW7 P5. Let A be a 2×2 complex matrix whose only eigenvalue is λ. Prove that its minimal polynomial is either $x - \lambda$ or $(x - \lambda)^2$. Show that the first case occurs exactly when $A = \lambda I$, and conclude that A is diagonalizable if and only if $A = \lambda I$.

Hint: Prop. 67 gives degree ≤ 2, Prop. 66 says the only root is λ, (T6) factors over $\mathbb{C}$; then F2 and Prop. 68.

HW7 P6. Consider

$$C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find its eigenvalues, eigenspaces, and minimal polynomial. Decide whether C is diagonalizable.

Hint: Method 2: eigenvalues from the diagonal (Prop. 60); test $(x - 2)(x - 3)$ first.

HW7 P7 (optional). Let T be an operator whose minimal polynomial splits as $m_T(x) = (x - \lambda_1) \cdots (x - \lambda_r)$ with distinct roots. For a fixed j, construct a polynomial p_j such that $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$, and explain why $p_j(T)$ is the projection onto $E(\lambda_j, T)$ along the sum of the other eigenspaces.

Hint: the polynomials g_j of Prop. 68 and F9.

M1. Computing minimal polynomials. For each matrix below, find the minimal polynomial and decide whether the matrix is diagonalizable (for P, decide over $\mathbb{R}$ and over $\mathbb{C}$).

$$\text{(a) } A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \text{(b) } P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \text{(c) } B = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

Hint: (a) Method 5 (rank one); (b) $P^3 = I$ and e_1 is a cyclic vector (F7(c)), then F8; (c) F4 on the diagonal blocks.

M2. Operators on polynomials. Let $n \geq 1$. (a) Let $T\colon \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$, $T(p) = xp'$. Find m_T and decide whether T is diagonalizable. (b) Let $S\colon \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$, $(Sp)(x) = p(x+1)$. Prove that $m_S = (x-1)^{n+1}$, and decide whether S is diagonalizable.

Hint: (a) $T(x^k) = kx^k$. (b) $S - I$ lowers the degree by at least 1, and $(S-I)^n x^n = n!$.

M3. Let V be a finite-dimensional real vector space and $T \in \mathcal{L}(V)$ with $T^3 = T$. Prove that T is diagonalizable and that every eigenvalue of T is -1, 0 or 1.

Hint: $x^3 - x = x(x-1)(x+1)$: the shortcut "Diagonalizable without finding eigenvalues", then Prop. 66.

M4. Let $V \neq \{0\}$ be finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $\deg m_T = 1$ if and only if $T = cI$ for some $c \in \mathbb{F}$.

Hint: F2.

M5. Minimal polynomial of the inverse. Let T be invertible with $m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0$. Prove that $m_{T^{-1}}(x) = \frac{1}{a_0} x^d m_T(1/x)$.

Hint: F6(c).

M6. Using the minimal polynomial to compute. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Find m_A, write A^{-1} as a polynomial in A, and write A^3 as $aA + bI$.

Hint: Method 6, then F1 for A^{-1}, then reduce A^3 with $A^2 = 5A + 2I$.

R1 (Midterm 2, Question 3). On $\mathbb{R}^3$, let T have standard matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(a) Determine its eigenvalues and corresponding eigenspaces. (b) Determine its minimal polynomial with justification. (c) Is T diagonalizable? Explain.

Hint: solve the equations $(A - 2I)v = 0$ and $(A + I)v = 0$; then test both $(x-2)(x+1)$ and $(x-2)^2(x+1)$. The roots alone do not justify the exponent.

R2 (Practice Midterm 2, Question 3). Let T act on a nonzero finite-dimensional complex space. You may use that T has an eigenvalue. Prove that T has an upper-triangular matrix, by induction: (a) handle dimension 1; (b) choose an eigenvalue λ, put $U = \operatorname{range}(T - \lambda I)$, and prove that U is invariant and $\dim U < \dim V$; (c) obtain a basis $u_1, \ldots, u_m$ of U with $Tu_j \in \operatorname{span}(u_1, \ldots, u_j)$; (d) extend it by $v_1, \ldots, v_r$ and use $Tv_j = (T - \lambda I)v_j + \lambda v_j$.

Hint: Prop. 61; include the case $U = \{0\}$.

R3 (Practice Midterm 2, Question 4). Let T act on a nonzero finite-dimensional real space. (a) If $m_T(0) \neq 0$, prove that T is invertible by expressing T^{-1} as a polynomial in T. (b) If $m_T(0) = 0$, prove that T is not invertible, using minimality. (c) For $R(x, y) = (-y, x)$ on $\mathbb{R}^2$, find m_R. (d) Use (a) to express R^{-1} in terms of R and check directly.

Hint: factor T out of every nonconstant term; in (b), an inverse would cancel a factor x and give a relation of smaller degree.

R4 (Midterm 2 review, Problem 4). (a) On the real space $V = \operatorname{span}(1, \sin x, \cos x, \sin 2x, \cos 2x)$, find the eigenvalues of differentiation and decide whether it is diagonalizable. (b) For $T(x, y) = (y, -x)$ on $\mathbb{C}^2$, find the eigenvalues and a basis giving an upper-triangular matrix. (c) Show that the same formula on $\mathbb{R}^2$ has no upper-triangular basis.

Hint: (a) write $Df = \lambda f$ in coordinates for the five spanning functions; (b), (c): $T^2 = -I$, and an upper-triangular basis starts with an eigenvector.

R5 (Midterm 2 review, Problem 6; LADR 5C, Exercise 7). Let $T \in \mathcal{L}(V)$ and $v \in V$. Prove that there is a unique monic polynomial p_v of least degree with $p_v(T)v = 0$, and that p_v divides m_T.

Hint: F7(a); for uniqueness, two such monic polynomials differ by one of smaller degree that is 0 on v.

R6 (Midterm 2 review, Problem 8: true or false). (a) Every vector space can be written as a direct sum of eigenspaces of some operator. (b) An operator on a real vector space always has an eigenvector. (c) The dimension of $E(\lambda, T)$ is the multiplicity of $x - \lambda$ in the minimal polynomial. (d) For an upper-triangular matrix, the minimal polynomial depends on the entries above the diagonal.

Hint: each statement has a 2×2 or 3×3 example in the Traps boxes; for (b) also see F11.

R7 (Additional Midterm Practice). Let $T \in \mathcal{L}(V)$ with $T^2 = T$. (a) Prove that $\operatorname{range} T \cap \operatorname{null} T = \{0\}$. (b) Prove that $V = \operatorname{range} T + \operatorname{null} T$. (c) If $S \in \mathcal{L}(V)$ and $ST = TS$, prove that $\operatorname{range} T$ and $\operatorname{null} T$ are invariant under S, so that S has a block diagonal matrix in a basis of $\operatorname{range} T$ followed by a basis of $\operatorname{null} T$.

Hint: (a), (b): write $v = Tv + (v - Tv)$; (c): the argument of (T5) with S in place of $p(T)$, then (T8).